TIC 405392578

Eclipsing binary candidate dossier - Star Vetter

CLEAN

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BLS EPHEMERIS
  Period       5.162568 d
  Epoch (t0)   1517.8065 BTJD
  Duration     0.1200 d
  Depth        3.388 %  (33,875 ppm)
  SNR          95.6

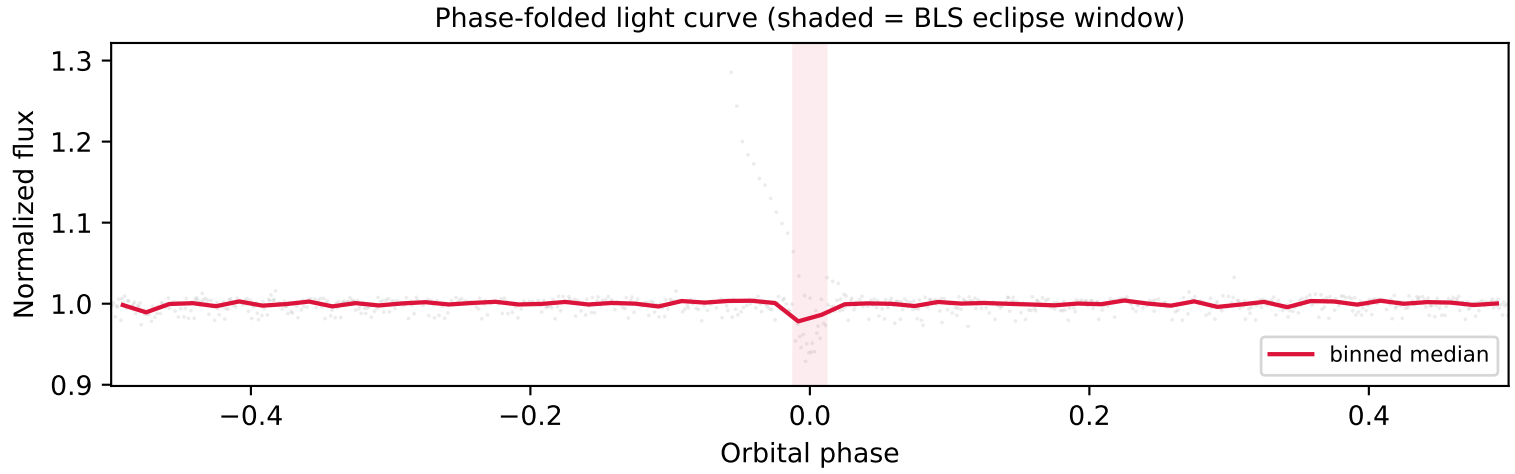
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     4  (gate >= 3)
  Robust depth ratio    1.08  (gate >= 0.50)
  Robust depth sigma    22.2  (gate >= 7.0)
  Odd/even z-score      0.72  (consistent)
  Secondary eclipse     not detected
```

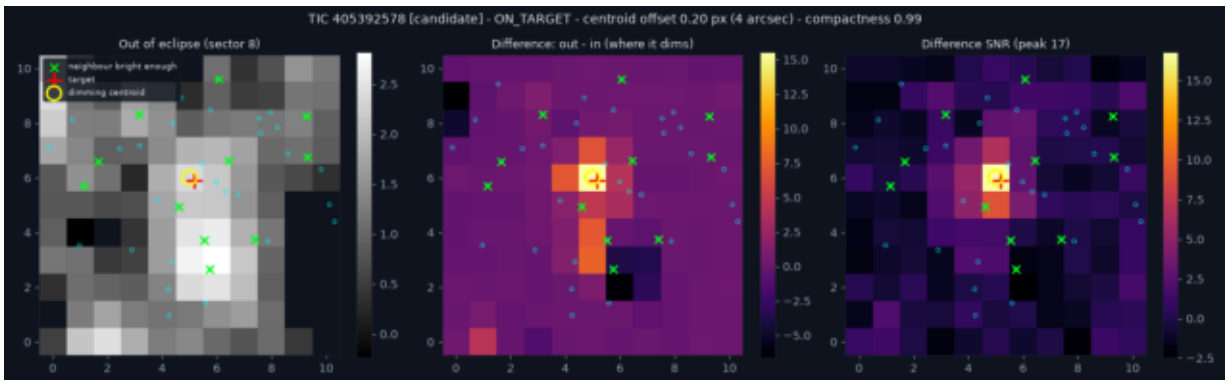
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 8    light-curve dip in this sector: 22.1 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   16.79   (gate >= 5.0)      Compactness   0.99   (gate >= 0.50)
  Dimming centroid offset     0.20 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 405392580
    1.13 px away, dTmag +1.87 - farther from the dimming than the target
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No caveats - clean detection on every check.